Chapter test

19 Homeostasis

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. In humans, homeostasis means that our body tries to

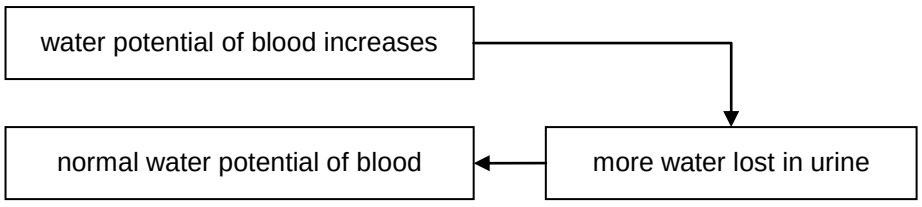
   A. keep the body temperature as high as possible to boost enzyme activities.
   B. keep the internal parameters unchanging.
   C. maintain a stable internal environment.
   D. adapt to changes in the external environment. ☐

2. Which of the following parameters of our body is / are under homeostatic control?

   (1) gas content in blood
   (2) glucose level in blood
   (3) fat reserves in cells

   A. (2) only
   B. (1) and (2) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

3. The diagram below refers to the control of water potential of the blood.



   Why is this an example of negative feedback mechanism?

   A. It maintains a constant water potential of the blood.
   B. It lowers the water potential of the blood.
   C. It increases the change in the water potential of the blood.
   D. It reverses the change in the water potential of the blood. ☐

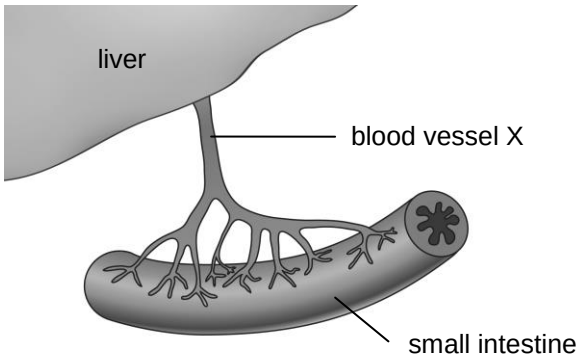
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4. Which of the following is / are involved in the regulation of the blood glucose level?

(1) pancreas
(2) liver
(3) medulla oblongata

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3) ☐

★ 5. The diagram below shows blood vessel X which carries digested food from the small intestine to the liver.



Which of the following combinations correctly describes the level of glucose in blood vessel X and the level of glycogen in the liver shortly after a meal rich in carbohydrates?

	Glucose in blood vessel X	***Glycogen in the liver***
A.	low	decreasing
B.	low	increasing
C.	high	decreasing
D.	high	increasing ☐

6. How does insulin affect glucose uptake by muscle cells and carbohydrate conversion by liver cells?

	Glucose uptake by muscle cells	***Carbohydrate conversion by liver cells***
A.	increases	glucose to glycogen
B.	increases	glycogen to glucose
C.	decreases	glucose to glycogen
D.	decreases	glycogen to glucose ☐

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7. Glucagon is secreted by the pancreas

(1) after a meal rich in carbohydrates.
(2) during physical exercise.
(3) during resting.

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

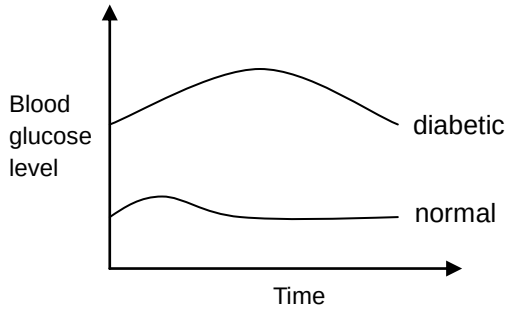
8. Which of the following statements about glucagon is / are correct?

(1) It is transported out of the pancreas through the pancreatic duct.
(2) It catalyses the conversion of stored glycogen to glucose in the liver.
(3) It raises the blood glucose level.

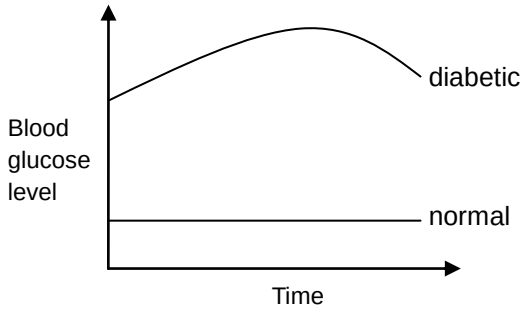
A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only ☐

★ 9. A healthy person and a diabetic patent drank the same volume of glucose solution. Which of the following graphs correctly shows the changes in their blood glucose level?

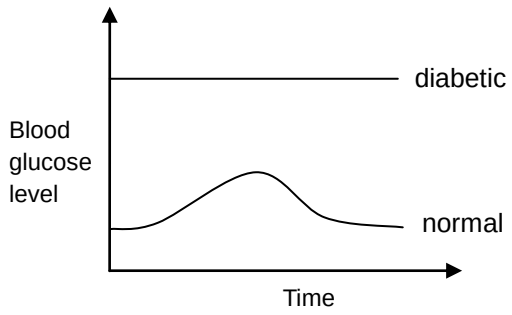
A.



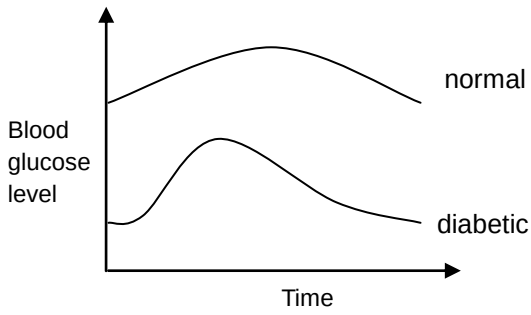
B.



C.



D.



☐

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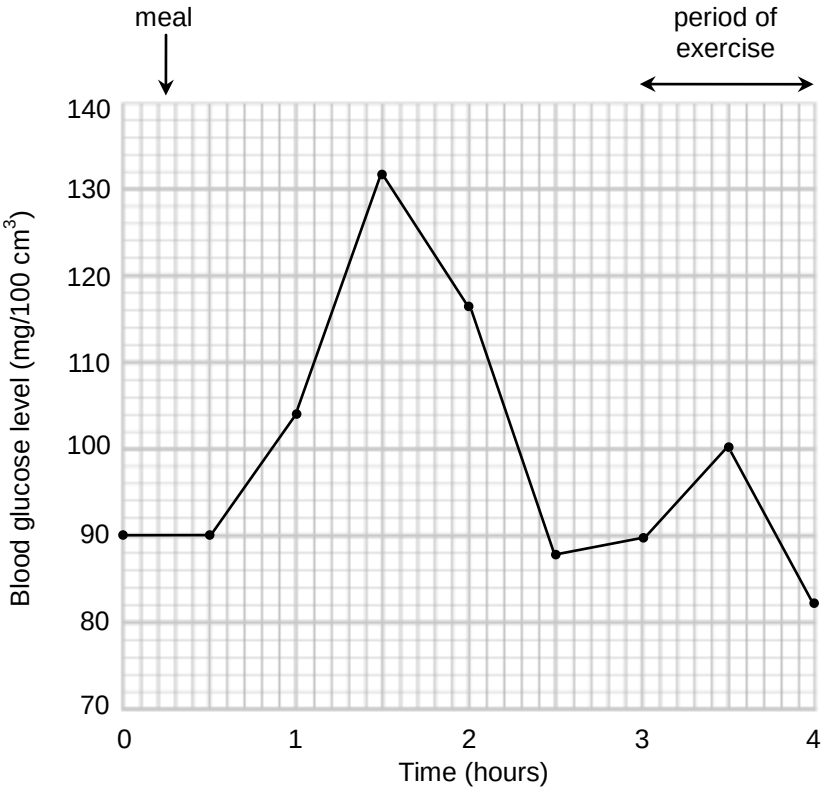
★ 10. Diabetic patients are advised to consume starch instead of simple sugars because

A. simple sugars will be excreted in urine.
B. their kidneys cannot reabsorb all the simple sugars.
C. the rise in blood glucose level is less rapid after the consumption of starch.
D. starch has a higher energy value than simple sugars. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	19.1	19.1	19.1	19.2	19.2	19.2	19.2	19.2	19.2	19.2

B. Problem solving (20 marks)

11. The graph below shows the changes in the blood glucose level of a person over a 4-hour period.



(a) What is the normal blood glucose level of this person? (1 mark)

_____

(b) Explain the increase in the blood glucose level after the meal. (2 marks)

_____

_____

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(c) Explain the change in the blood glucose level between hour 1.5 and hour 2.5. (5 marks)

(d) (i) Which organ in the body provided the additional glucose to the blood between hour 3 and hour 3.5? What process in this organ caused this increase in the blood glucose level? (2 marks)

(ii) Explain the significance of such an increase in the blood glucose level during exercise. (2 marks)

(Total: 12 marks)

★★ 12. Peter is a diabetic. He needs injections of insulin to control his blood glucose level.

(a) Explain why insulin has to be taken in the form of injections rather than taken in orally? (2 marks)

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Peter measures his blood glucose level several times a day. According to his blood glucose level measured, Peter will adjust the dose of insulin to be injected before each meal. He tries to keep his blood glucose level within the normal range of 4.0 to 7.0 mmol/L. The table below shows his measurements over three days.

	Blood glucose level (mmol/L)						
	Before breakfast	**2 hours after breakfast**	**Before lunch**	**2 hours after lunch**	**Before dinner**	**2 hours after dinner**	**Before bedtime**
21 May	7.0	7.5	5.9	6.8	4.2	13.8	12.9
22 May	9.0	6.2	5.3	7.6	6.1	8.9	7.7
23 May	5.4	7.9	7.3		2.9	7.4	7.0

(b) Suggest a reason for

(i) the high blood glucose level 2 hours after dinner on 21 May. (1 mark)

(ii) the low blood glucose level before dinner on 23 May. (1 mark)

(c) Suggest why Peter needs to measure his blood glucose before bedtime. (2 marks)

(d) (i) Insulin injection does not work for some diabetics. Explain why. (1 mark)

(ii) Suggest ***one*** way that may help the diabetics in (i) to control their blood glucose level. (1 mark)

(Total: 8 marks)

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C. Exam challenge (10 marks)

★★ 13. In a blood glucose test, two men (X and Y) were starved for 12 hours. They were then given an equal volume of the same glucose solution to drink. Their blood glucose levels were measured immediately afterwards and at 30-minute intervals for four hours. The results are shown in the table below.

	Blood glucose level (mg/100 cm^3)	
Time after intake of glucose solution (min)	**X**	**Y**
0	80	120
30	100	180
60	150	280
90	120	270
120	85	250
150	80	225
180	80	200
210	85	140
240	90	120

(a) Why is it necessary for the two men to starve for 12 hours before taking the blood glucose test? (2 marks)

(b) Which of X and Y is more likely to be a diabetic? Give ***two*** pieces of evidence to support your answer. (3 marks)

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(c) In view of your answer to (b), explain the difference in the blood glucose level of X and Y between 30 and 60 minutes. (4 marks)

(d) What caused the blood glucose level to decrease in the diabetic? (1 mark)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	B	3.	D	4.	C	5.	D
6.	A	7.	D	8.	B	9.	A	10.	C

(1 mark each)

B. Problem solving (20 marks)

11. (a) 90 mg/100 cm^3 (1)

(b) The starch in the meal is digested into glucose (1)
and absorbed into the blood in the small intestine. (1)

(c) The pancreas detected an increase in the blood glucose level. (1)
More insulin was secreted by the pancreas into the blood. (1)
Insulin stimulated liver cells to convert the excess glucose to glycogen for storage. (1)
It also stimulated body cells to take up more glucose from the blood. (1)
As a result, the blood glucose level decreased and returned to normal. (1)

(d) (i) Liver (1)
Conversion of stored glycogen to glucose (1)

(ii) This provides more glucose for the respiration of muscle fibres (1)
to release more energy for muscle contraction during exercise. (1)

(Total: 12 marks)

12. (a) Insulin is a protein. (1)
If it is taken orally, it will be broken down / digested by the digestive enzymes in the stomach. (1)

(b) (i) He injected too little insulin before dinner. / He forgot to inject insulin. / His dinner contained too much carbohydrate (sugar). (1)

(ii) He injected too much insulin before lunch. / He didn't eat at lunch time. / He ate too little carbohydrate in the afternoon. / He exercised in the afternoon. (1)

(c) To ensure the blood glucose level is within a safe range. (1)
As the blood glucose level continues to fall, he may faint during the night. (1)

(d) (i) In some diabetics, their body cells do not respond to insulin. (1)

(ii) Exercising regularly / Adopting low-sugar diets (1)

(Total: 8 marks)

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C. Exam challenge (10 marks)

13. (a) Food taken in may increase the blood glucose level (1)
and interfere with the results of the test. (1)

(b) Y (1)
Any ***two*** of the following reasons: (2)
- His initial blood glucose level was higher than X. (1)
- His blood glucose increased more rapidly / increased to a higher level after the intake of glucose solution. (1)
- It took a longer time for his blood glucose to return to the original level. (1)

(c) In X, the initial increase in the blood glucose level stimulated the pancreas (1)
to secrete more insulin into the blood, (1)
while there was no / less insulin secretion in Y. (1)
Insulin stimulates the conversion of glucose into glycogen in the liver / the update of glucose by body cells, (1)
so the increase in blood glucose level in X was smaller.

(d) It was caused by the loss of glucose in urine. (1)

(Total: 10 marks)

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